Phase 3 – Intrusion Detection Battle (Suricata)

1.Installing Suricata –

- sudo apt update && sudo apt install suricata -y
- suricata -V
- sudo systemctl start suricata

2. Custom Rules File (/etc/suricata/rules/local.rules)

These rules look for specific patterns in the packet headers and payloads.

- Rule 1: ICMP Recon (Alert on Ping)

alert icmp any any -> any any (msg:"ICMP Ping Detected"; sid:1000001; rev:1;)

- Rule 2: Nmap SYN Scan Detection

alert tcp any any -> any any (msg:"Nmap SYN Scan Detected"; flags:S; ack:0; sid:1000002; rev:1;)

- Rule 3: Telnet Usage

alert tcp any any -> any 23 (msg:"Telnet Traffic Detected"; sid:1000003; rev:1;)

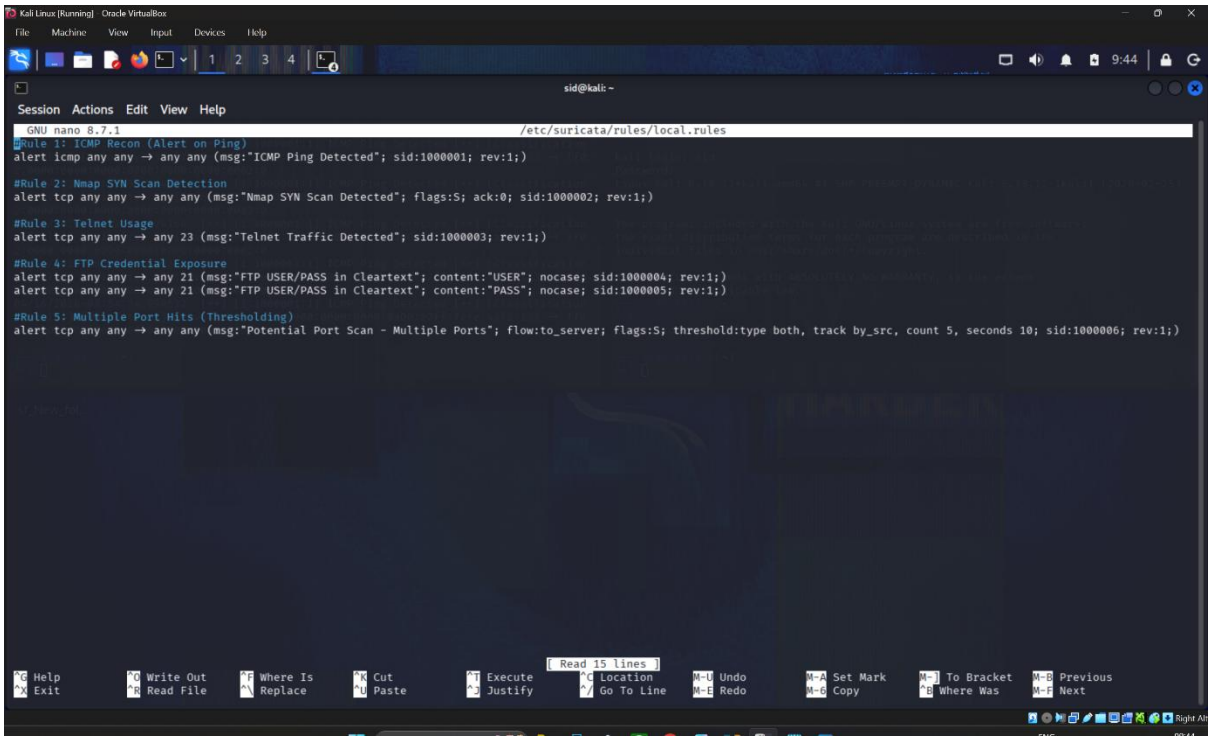
- Rule 4: FTP Credential Exposure

alert tcp any any -> any 21 (msg:"FTP USER/PASS in Cleartext"; content:"USER"; nocase; sid:1000004; rev:1;)

alert tcp any any -> any 21 (msg:"FTP USER/PASS in Cleartext"; content:"PASS"; nocase; sid:1000005; rev:1;)

- Rule 5: Multiple Port Hits (Thresholding)

alert tcp any any -> any any (msg:"Potential Port Scan - Multiple Ports"; flow:to_server; flags:S; threshold:type both, track by_src, count 5, seconds 10; sid:1000006; rev:1;)



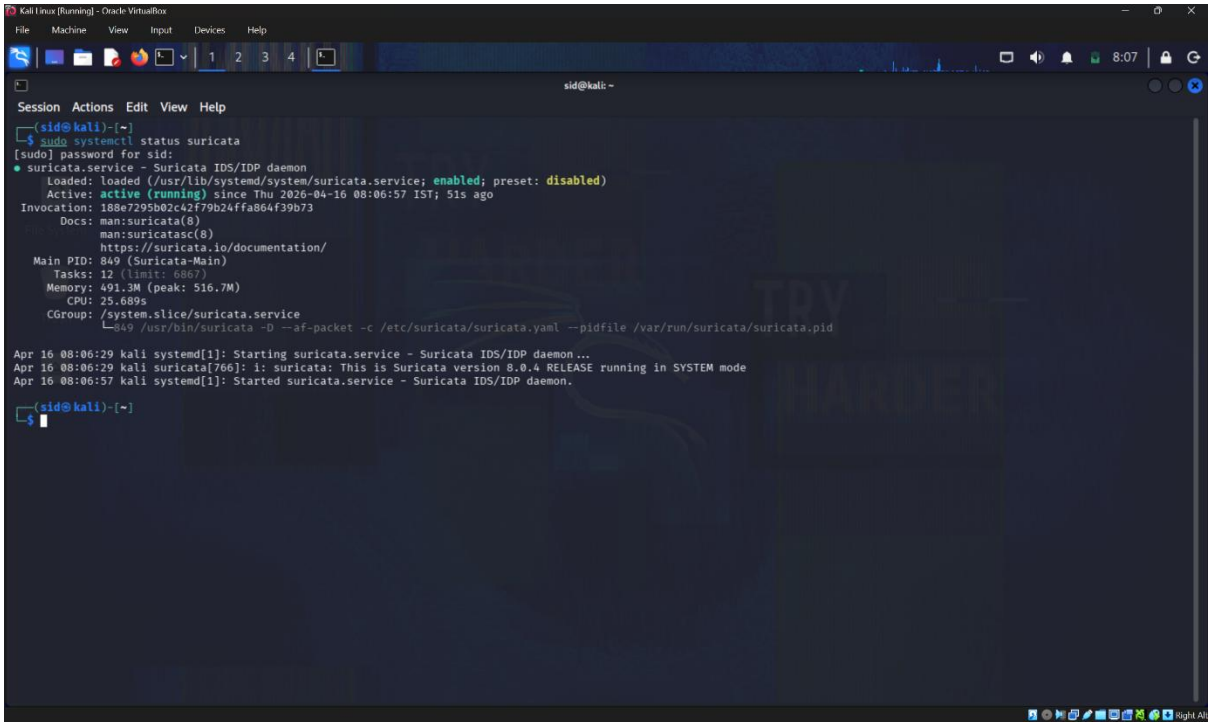
3. Commands Used for Testing

- The Attacker, execute the following to trigger the rules:

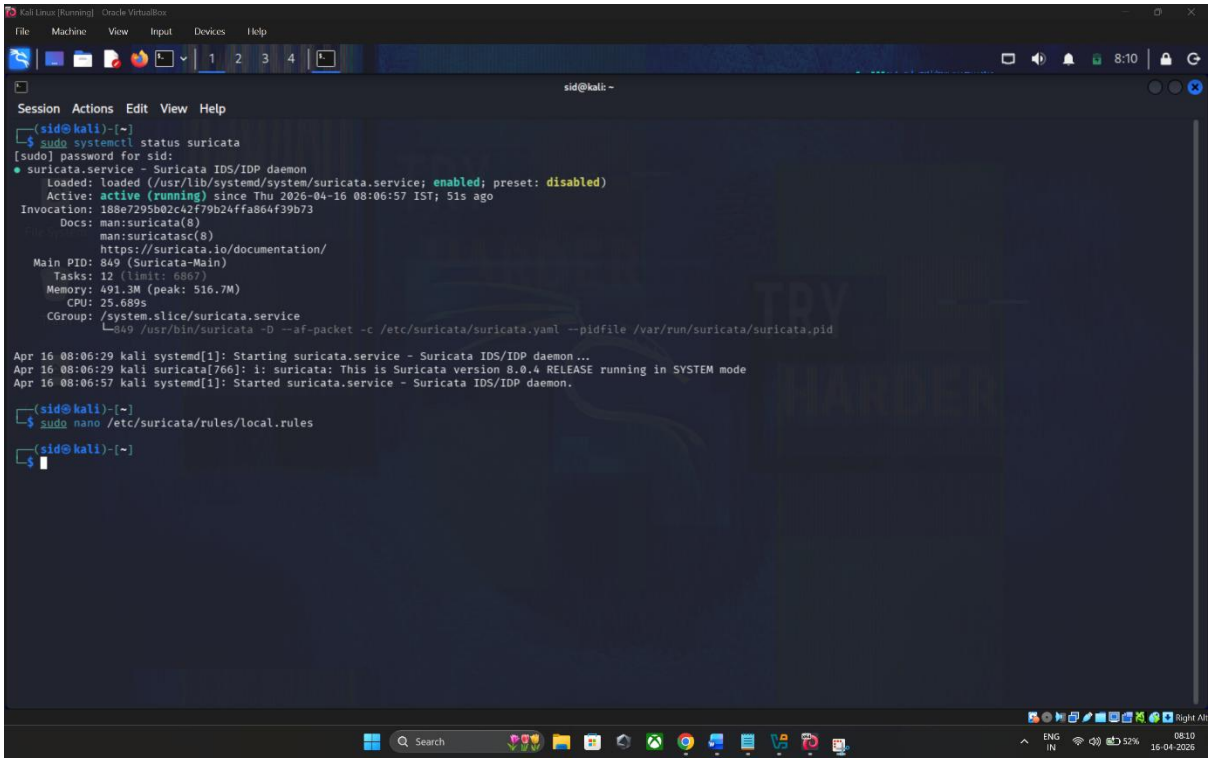
Task	Command	Target IP
Ping Test	ping 10.0.2.3	Kali IP
Nmap Scan	nmap -sS 10.0.2.3	Kali IP
Telnet	telnet 10.0.2.3	Kali IP
FTP Login	ftp 10.0.2.3	Kali IP

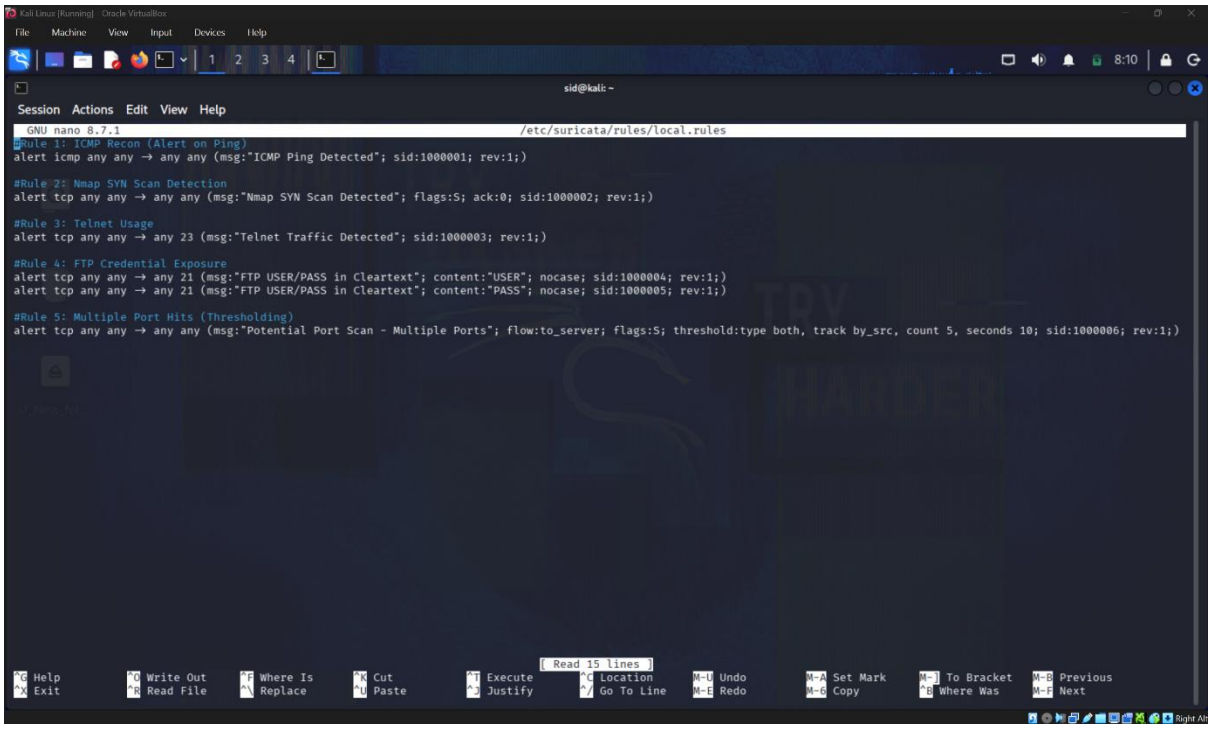
4. Proof of work

- Check status of Suricata

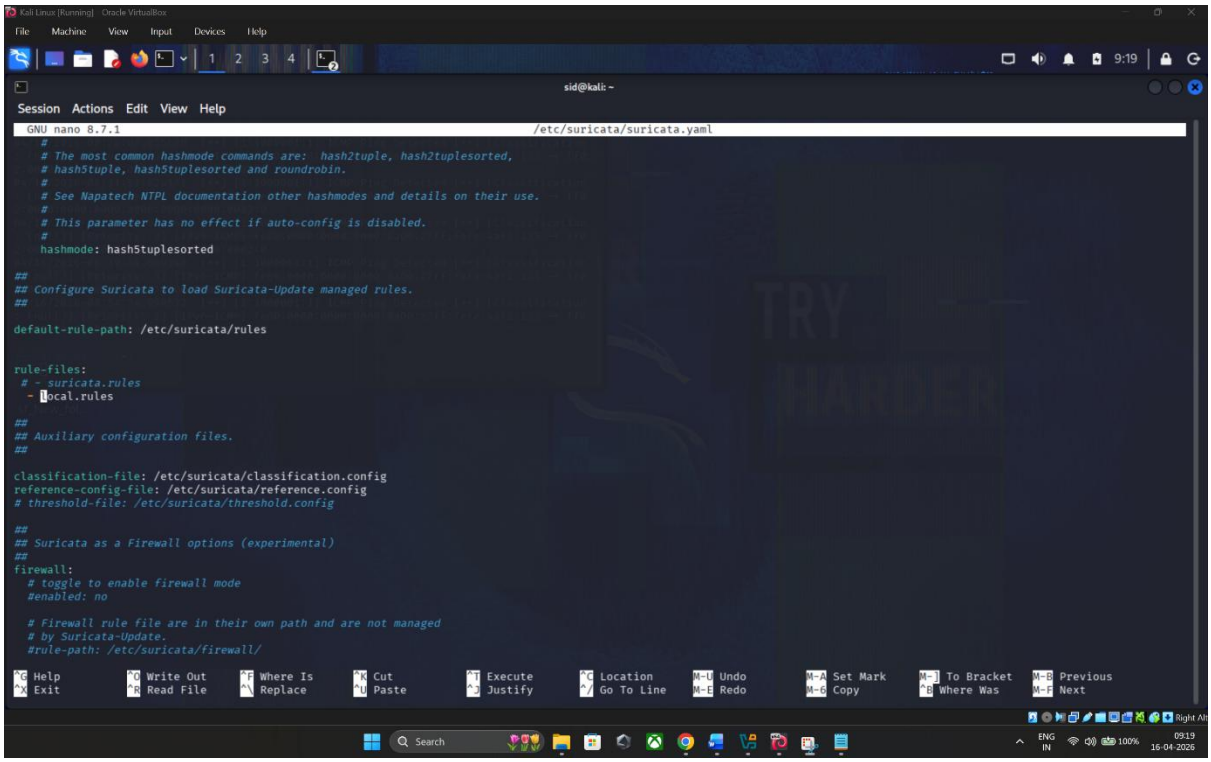


- Creating new rule file and adding custom rules –

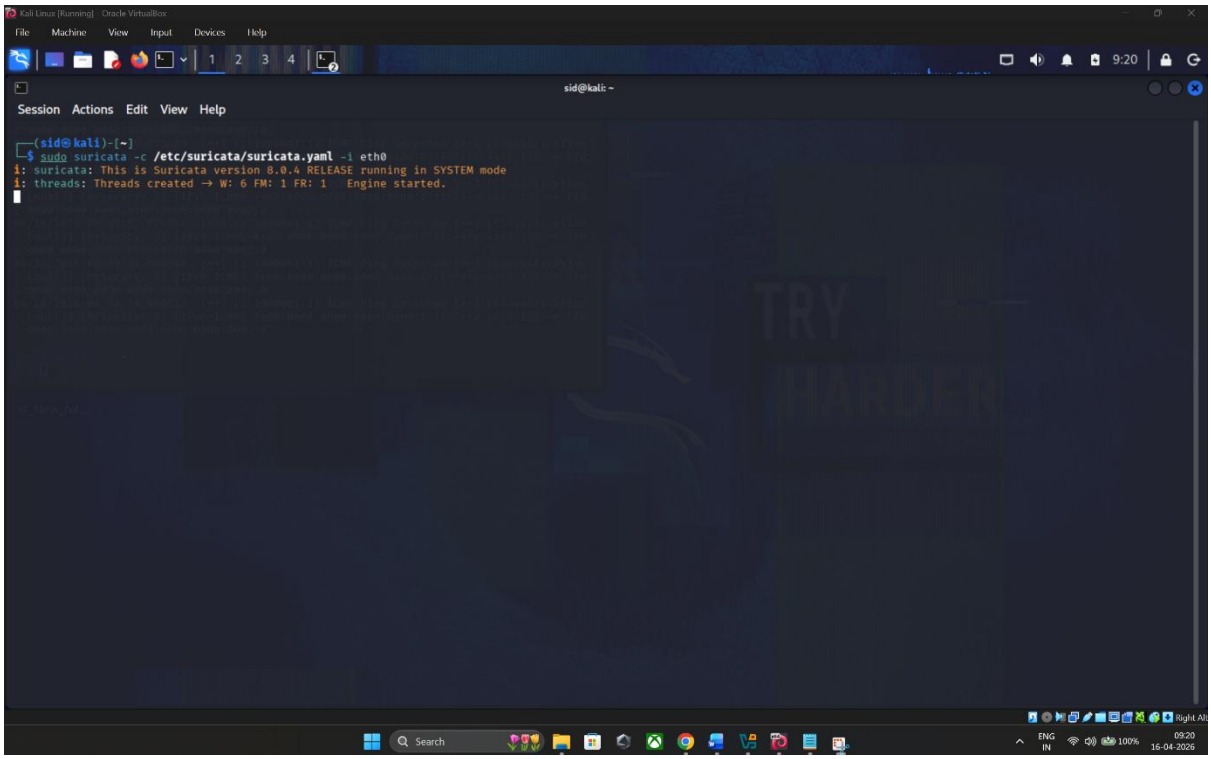




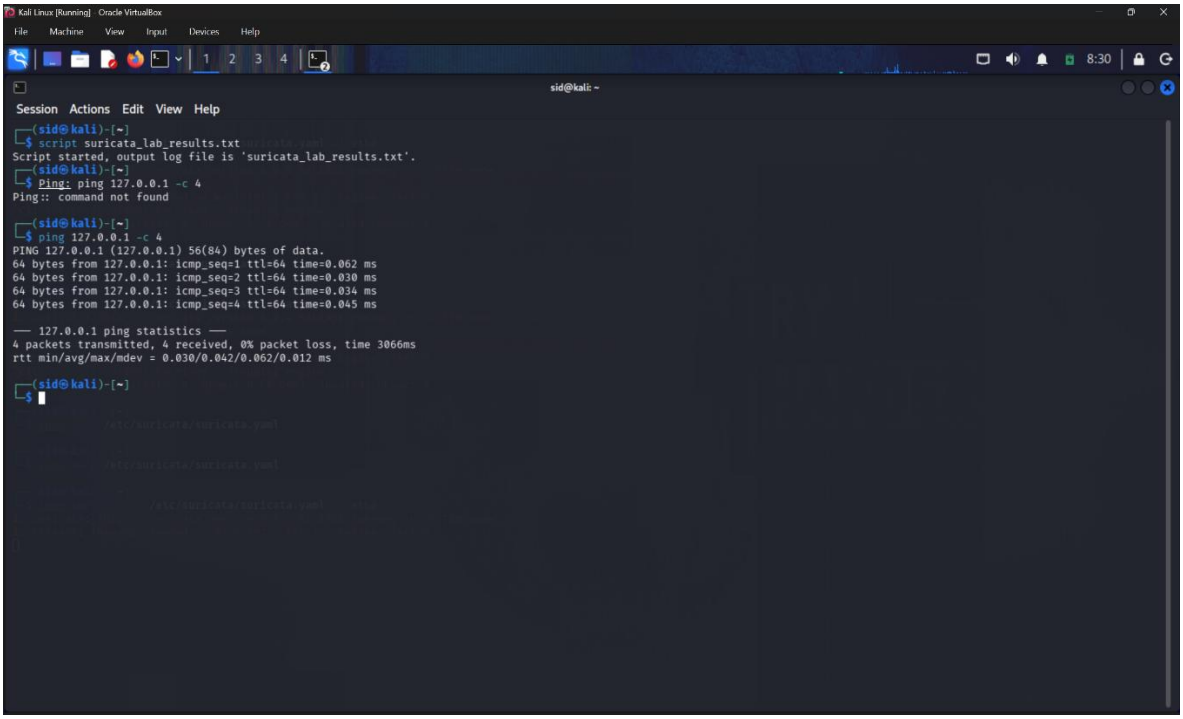
- Enabling custom rules to Suricata Config –

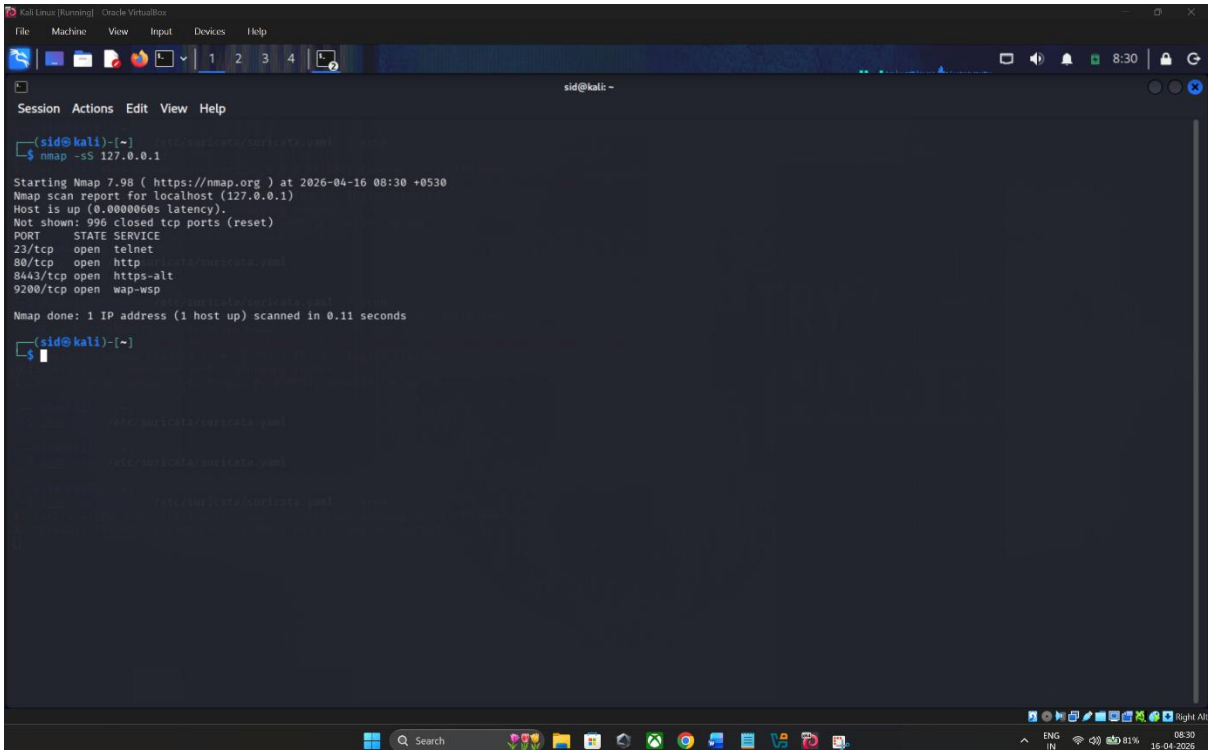


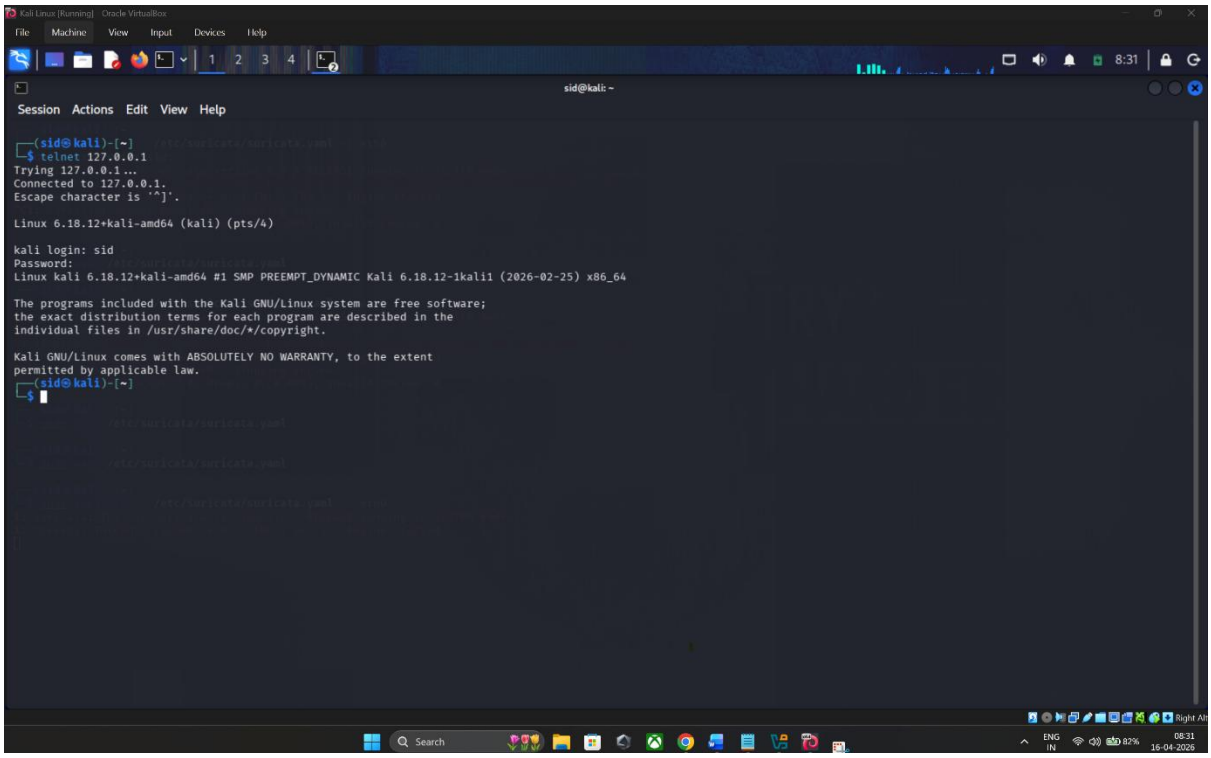
- Starting Suricata service



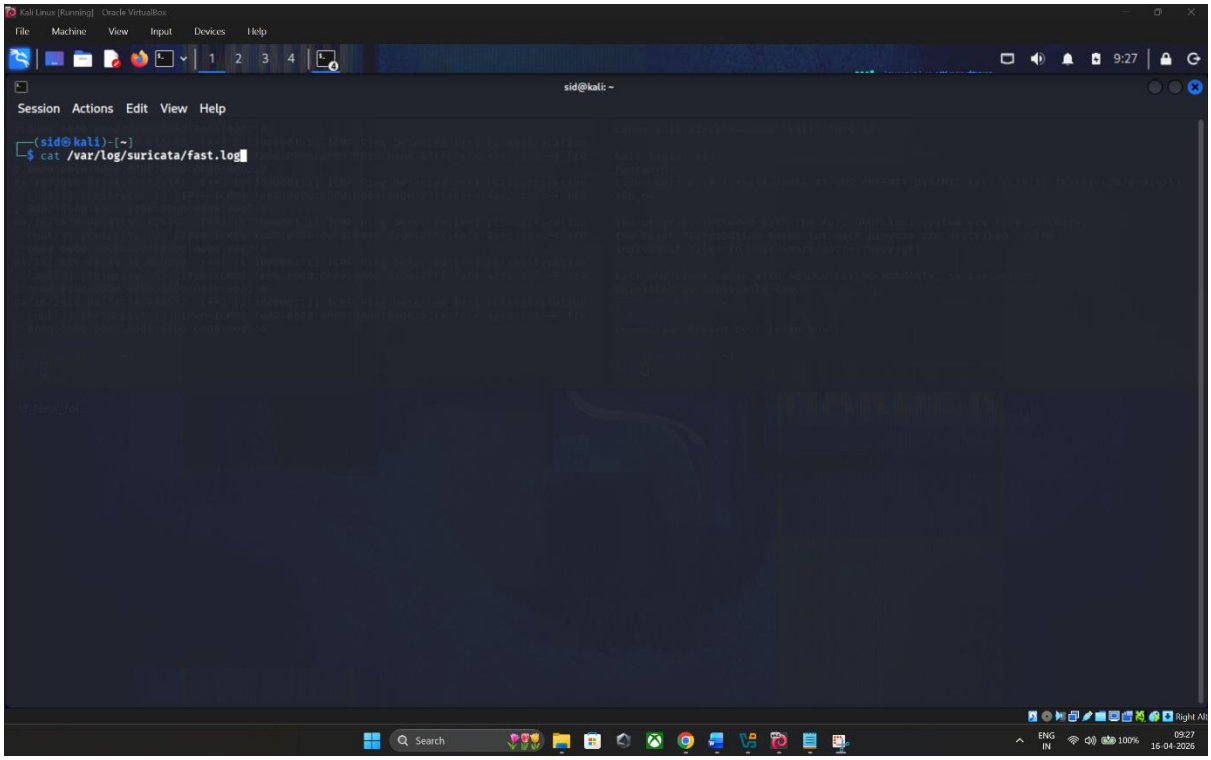
- Testing Ping, nmap and telnet

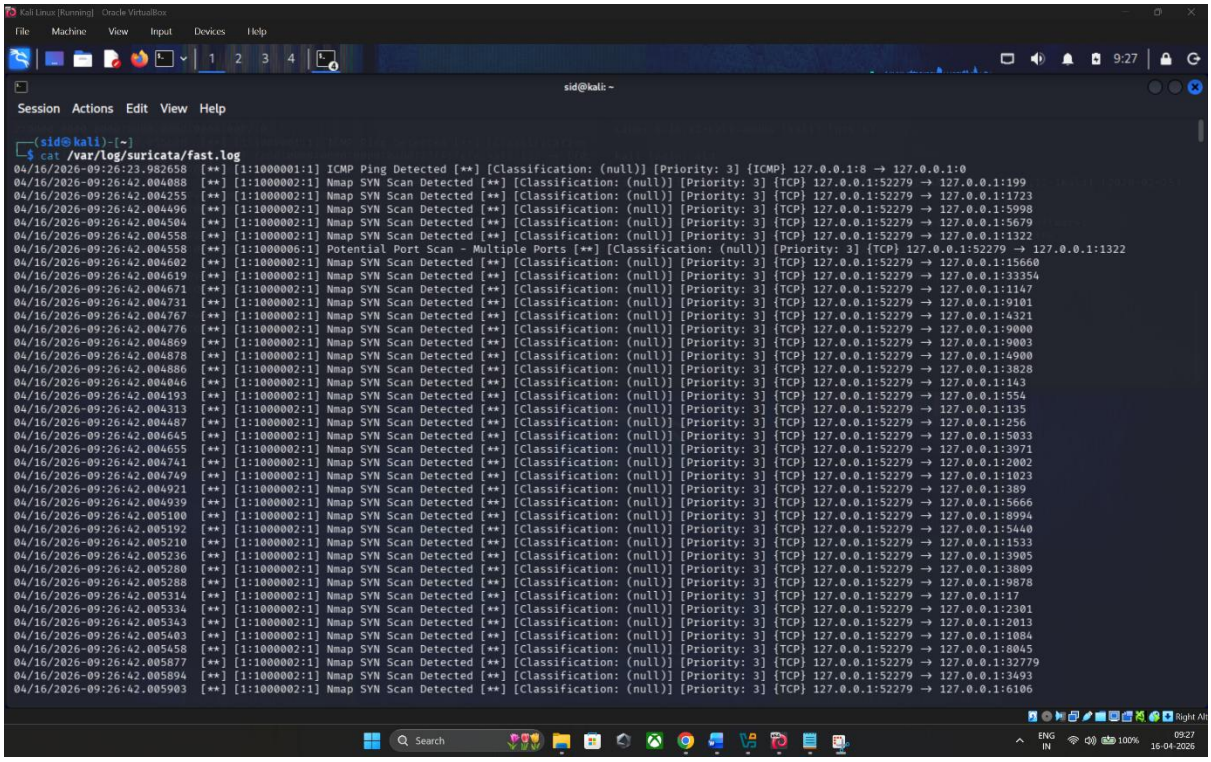



Kali Linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
sid@kali: ~
Session Actions Edit View Help
┌──(sid㉿kali)-[~]
└─$ nmap -sS 127.0.0.1
Starting Nmap 7.98 (https://nmap.org) at 2026-04-16 08:30 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000060s latency).
Not shown: 996 closed tcp ports (reset)
PORT STATE SERVICE
23/tcp open telnet
80/tcp open http
8443/tcp open https-alt
9200/tcp open wap-wsp
Nmap done: 1 IP address (1 host up) scanned in 0.11 seconds
┌──(sid㉿kali)-[~]
└─$


Kali Linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
sid@kali: ~
Session Actions Edit View Help
┌──(sid㉿kali)-[~]
└─$ telnet 127.0.0.1
Trying 127.0.0.1 ...
Connected to 127.0.0.1.
Escape character is '^]'.
Linux 6.18.12+kali-amd64 (kali) (pts/4)
kali login: sid
Password:
Linux kali 6.18.12+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.18.12-1kali1 (2026-02-25) x86_64
The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
┌──(sid㉿kali)-[~]
└─$

- Checking recorded log on **fast.log**





5. WHICH RULE TAKES PRIOTIRY

- **Rule 4** (FTP Credential Exposure) is the highest priority.

Priority	Rule	Description	Reason
1 (High)	**Rule 4**	**FTP Credentials**	**Critical:** Actual loss of credentials. Direct path to system compromise.
2 (Med)	**Rule 3**	**Telnet Usage**	**Warning:** Use of an insecure protocol that *will* lead to credential theft if used.
3 (Low)	**Rule 2 & 5**	**Nmap/Port Scans**	**Informational:** Active probing. Hostile intent is clear, but no breach has occurred.
4 (Lowest)	**Rule 1**	**ICMP Ping**	**Noise:** Ping is often used for legitimate network troubleshooting.